An audited benchmark for long-tailed short-text classification in Portuguese: 250k retail product descriptions, 621 categories, LLM annotations, and a reproducible active-learning library

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Draft v0.1 — July 19, 2026

Abstract

Public benchmarks for product-title classification are scarce, rarely in Portuguese, and almost never ship with an audit of their own label noise. We consolidate, audit and instrument a corpus of ≈250,000 short retail product descriptions from Brazilian fiscal receipts (4–50 characters, uppercase, aggressive abbreviation), labeled over a closed catalog of 621 product-type categories with severe long-tail imbalance. The raw corpus has been publicly available on Kaggle since 2022 (DOI `10.34740/kaggle/dsv/4265348`); this paper adds the three artifacts that turn it into a *measurement-grade* resource rather than a raw dump: (i) a **label-noise audit**: a census of conflicting labels (identical normalized text, divergent gold), a multi-gold re-scoring protocol, and the resulting measurement ceiling (≈99.3%), shown to shift oracle accuracies by at most 0.7 points without reordering any comparison; (ii) **paired evaluation samples** (random $n = 1{,}000$; stratified 3-per-class $n = 1{,}863$) with zero-shot annotations from six LLM oracles under a recorded (model, provider, contract, batch, temperature, date) provenance tuple — enabling instrument-level replication studies; and (iii) an open **active-learning library** (hexagonal architecture; prototype, linear and transformer classifiers; uncertainty, density and stratified selection policies; LLM oracle adapters with strict output validation, annotation caching and cost metering) plus a visual pipeline builder, with which every published number is reproducible from versioned experiment states. Baselines span a per-class prototype model (89.6% accuracy at full supervision), fine-tuned BERTimbau, and the LLM-oracle plateau (77–83% zero-shot). The resource targets research on label-efficient learning, LLM annotation, and evaluation methodology in lower-resourced languages.

Keywords: language resources; Portuguese; short text; product classification; label noise; active learning; reproducibility.

1 Motivation

Three research communities need this resource and currently improvise. Work on *LLM annotation* needs closed-taxonomy tasks at realistic granularity with audited gold — published product studies stop at tens to ≈370 categories (Roumeliotis et al., 2025; Gholamian et al., 2024), and public product corpora with real noise exist mainly in English or multilingual marketplace settings (Rączkowska et al., 2024; Karl and Scherp, 2023). Work on *label-efficient learning* (active learning, semi-supervision) needs pools large enough for full-trajectory experiments with a reserved population. Work on *Portuguese* NLP for commerce exists at scale only behind institutional walls (Machado and Veloso, 2026). This resource serves the three at once, and adds what none of the cited resources provides: an audit of its own gold standard and per-annotation LLM provenance.

2 Corpus

≈250k product descriptions as typed into Brazilian point-of-sale systems and printed on fiscal receipts: uppercase, 4–50 characters, domain abbreviations (`CERV` for beer, `LT` for can), brand and packaging tokens interleaved with type information. Labels: 621 product-type categories (e.g., *cerveja*, *enxaguante bucal*), assigned by a human back-office process over years of operation (Darú et al., 2022; Darú, 2024). After per-class filtering ($\geq$ 2 instances) and text-normalized deduplication the experimental view has 231,490 unique texts across 649 raw classes. The distribution is heavily long-tailed — the property that makes the benchmark hard: macro metrics are governed by hundreds of small classes.

3 The label-noise audit

Real gold standards err; a resource that hides it exports silent bias. We ship: (i) the **conflict census** — groups of identical normalized texts carrying divergent labels, with counts and examples [TODO: inserir estatísticas exatas do censo (nº de grupos, % de instâncias afetadas) a partir do artefato da auditoria]; (ii) a **multi-gold protocol** — re-scoring any system accepting any of the conflicting labels as correct; (iii) the **sensitivity result**: across six LLM oracles, excluding conflicts or applying multi-gold shifts accuracy by at most +0.7 points and never reorders systems — so the corpus supports comparative research despite its measured ≈0.7% gold-noise floor, and the measurement ceiling (≈99.3%) should accompany any absolute claim.

4 Paired LLM annotation samples

Two evaluation samples, annotated by six oracles (three cost regimes: commercial APIs, MaaS platforms, free tier) at temperature 0 with per-annotation provenance: **S-rand** (n = 1,000, production distribution) and **S-strat** (3 per class, n = 1,863, tail support). Every annotation records model, serving provider, output contract (enum / json-prompt / free), batch size, token counts and date — the metadata that companion work shows can move results more than model choice. These samples support replication of instrument-effect studies (output-contract artifact; serving-provider divergence) without any API spending.

5 Library and pipeline builder

The `activelearning` library (Python, hexagonal architecture — domain core isolated from adapters) provides: classifiers (per-class prototype PVBin, whose full-supervision baseline is 89.6% accuracy / 0.70 macro-F1; logistic SGD; BERTimbau fine-tuning (Souza et al., 2020)); selection policies (entropy, margin, least confidence, random, density-representative DRI-SL, predictionguided and prediction-stratified variants); oracle adapters (OpenAI-compatible providers, structured-output and prompted-JSON contracts, strict label validation, persistent annotation cache, cost metering); loop orchestration with stagnation-based stopping and state-file resumption; and a web pipeline builder (FlowBuilder) for no-code experiment assembly. Every experiment in the companion papers resolves to a versioned state under `experiments/`, and a Docker image plus a Colab notebook reproduce the GPU-dependent baselines.

6 Availability, licensing and ethics

Code and library: MIT license (public). Evaluation samples with LLM annotations and all per-annotation artifacts: released with this paper. Full corpus: publicly available on Kaggle since 2022 under a persistent identifier (DOI `10.34740/kaggle/dsv/4265348` (Darú, 2022)), [TODO: BLOQUEANTE: a página do Kaggle está com License = Unknown — definir licença explícita (recomendação: CC BY 4.0) na página do dataset antes da submissão; atualizar esta frase com a licença escolhida]. The audit artifacts released here reference the published version; [TODO: registrar hash/versão do CSV usado na tese vs. versão publicada (diferença esperada: correção 250.365 -> 250.221 itens — documentar como changelog ou nova versão no Kaggle)]. Texts are product descriptions only; no personal, transactional or merchant-identifying data is present.

7 Baseline summary

Full supervision: PVBin 89.6% / 0.70 macro-F1 (250k labels); BERTimbau 88.3% / 0.451 macro-F1 fine-tuned on a 50k pool (3 epochs, 32-token context, reserved-population evaluation — a deliberately economical uniform configuration; see the library for the exact recipe). Zero-shot LLM plateau: 77–83% accuracy, macro-F1 up to 0.80 (exceeding the light supervised baseline on the tail). Label-efficient reference points: entropy-based selection reaches 95% of its ceiling with 16–38% of pool labels depending on the classifier. These anchors let new methods report gains against published, artifact-backed numbers.

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